

Simulation and Modeling

Lecture 01 **Introduction**

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Course Objectives

- To be able to analyze systems, e.g., discrete event systems, for performance comparison and measurement
- To be able to build the model of systems
 - Model for both simulation and analytical analysis
- To be able to develop from the model
 - Simulation programs
 - Analytical Solutions
- To be able to analyze the output
 - From simulations and analytical solutions
 - Understand the desired characteristics of the target system

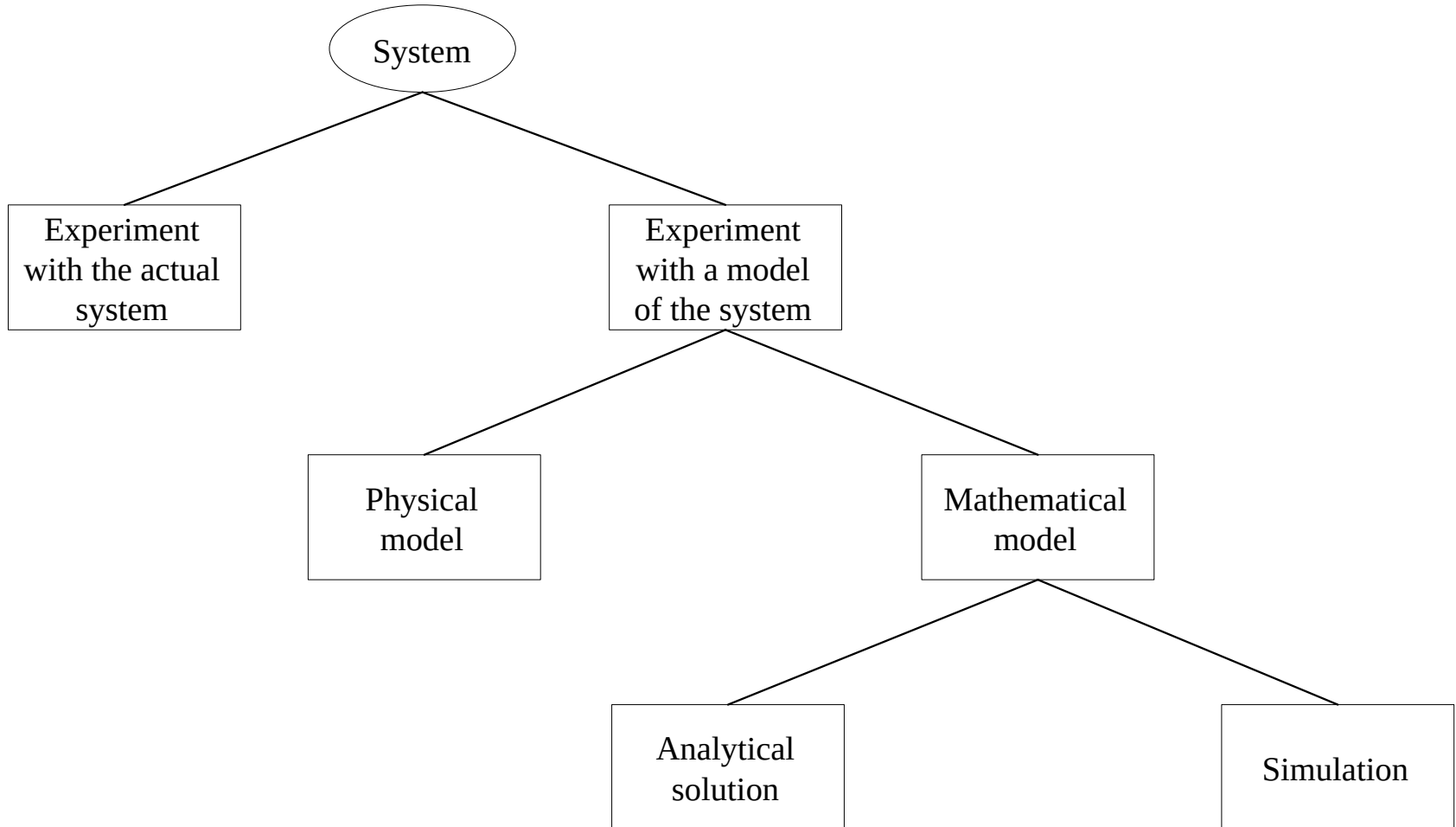
Text Books

- Simulation Modeling and Analysis
 - Averill M Law
- Discrete-Event Simulation: A First Course
 - C G Cassandras, and Stephene Lafortune
- Discrete-Event System Simulation
 - J Banks, JS Carson, BL Nelson, and DM Nicol
- Introduction to Discrete Event Systems
 - Lawrence Leemis, and Steve Park
- Queueing Theory

Simulation, Modeling and Analysis

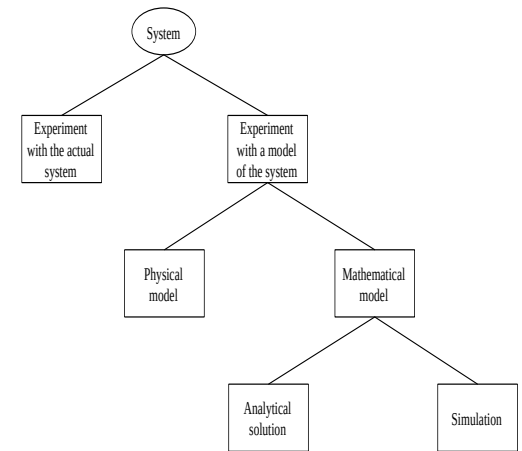
- Simulation
 - Imitation of the operation of a system or process
 - Generate an artificial history of running the system
- Modeling
 - An Alternate representation of the system physically or mathematically
- Performance Analysis
 - Understanding the behavior of the system
 - By analyzing the artificial history from simulation
 - By solving the model analytically
 - Inference of the operational characteristics

Analysis of Systems



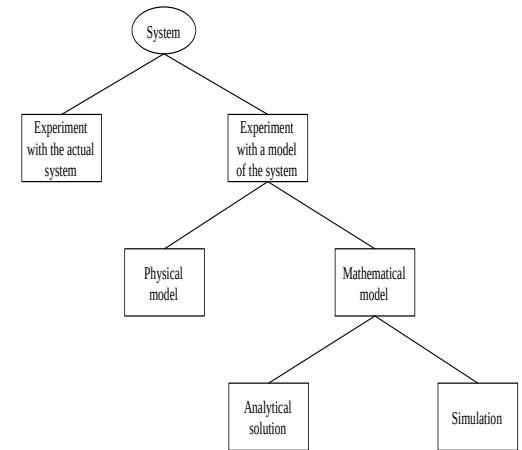
Experiment with Actual System/Model

- Actual System
 - Best to experiment with actual system
 - No question of validity
 - Rarely feasible to work
 - Too costly or too disruptive
 - E.g., reducing the # of tellers of a bank
 - Actual system might not exist
 - Behavior of proposed alternatives
 - System might not be accessible
 - Vulnerable to work with
 - Too secured



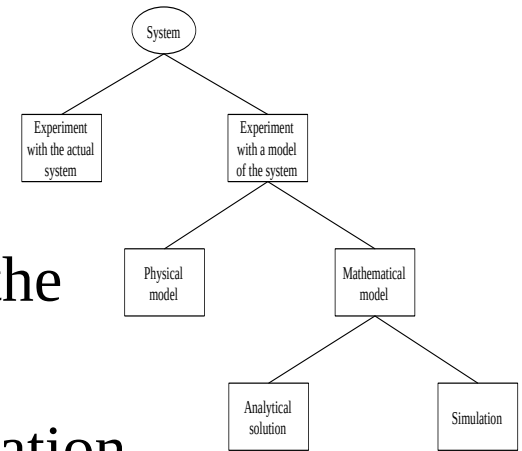
Experiment with Actual System/Model (Contd.)

- Model of a system
 - A representation of the real system
 - Usually, real systems are too complex
 - Difficult to represent
 - A # of assumptions/simplifications are made
 - Assumptions should be reasonable
 - Otherwise, the model might not represent the target system
 - Validation of the model is required



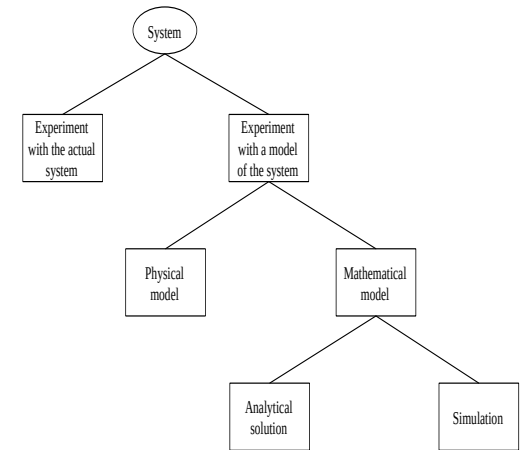
Physical Model/Mathematical Model

- Physical Model
 - A replica of the real system
 - At least the necessary components of the system
 - Components associated with the evaluation
 - Cockpit of a airplane used for pilot training
 - Might not suitable for many systems
 - A communication device is too costly to build
 - An IC cannot be made just for testing
 - Though, a fast food restaurant might be replicated in a warehouse



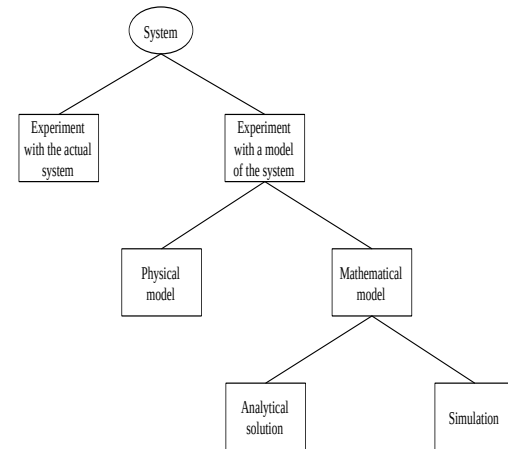
Physical Model/Mathematical Model (Contd.)

- Mathematical Model
 - Represent a system by logical components and their relationship
 - A set of variables might represent the component
 - A set of equations represent the relationship
 - Simplest example $d = rt$, where r represents the velocity, and t represents the time, and d represents the travelled distance
 - Solution of the mathematical model produces the desired characteristics of the system



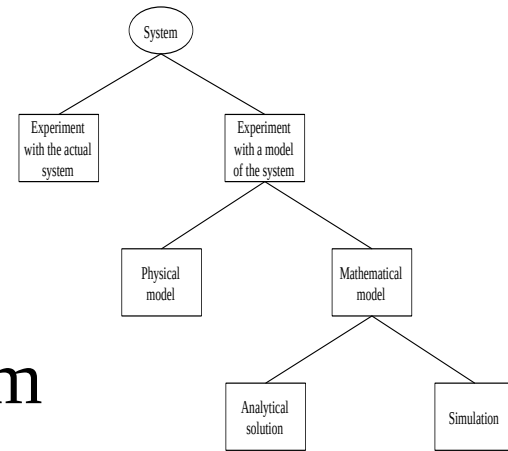
Analytical Solution/Simulation

- Analytical Solution
 - Quantity of interest are found by analytically solving the model
 - Easy for simple model
 - $t = d/v$
 - Complex system might require vast computing resources
 - A close-form solution might not exist
 - If analytical solution is available and computationally efficient
 - preferred over simulation



Analytical Solution/Simulation (Contd.)

- Simulation
 - Numerical techniques are used to evaluate the system
 - Imitate the behavior of the real system
 - For a set of inputs, a set of outputs are generated
 - System behavior is studied from the output data
 - Difficult to distinguish whether an observation represents the system interrelationship or just a random result



When Simulation is Appropriate and Not Appropriate

- To know the internal interaction of a complex system
- To know the information, organizational and environmental changes
- To experiment with new systems
 - proposed extension of a system
- To verify the analytical solution

- The problem can be solved by common sense
- The problem can be solved analytically
- Direct experiment can be easily performed
- resources and time are not available
- Sufficient data are not available
- System's behavior is too complex

Advantages and Disadvantages of Simulation

- Explore the following without disrupting the systems
 - New policies, operating procedures
 - Decision rules, information rules
- Following can be tested
 - New hardware design, physical layout
 - Transportation systems
 - Communications protocols and systems
- Hypothesis about how or why certain phenomena occur can be understood
- ‘What if’ questions about a system can be answered
 - Especially for new systems

- Model building requires special training
 - Two models of same system unlikely to be same
 - Packages with model are developed by the vendors
- Simulation results can be difficult to interpret
 - Simulation package includes output data analyzer
- Modeling and analysis can be time consuming
 - Advances in hardware and simulation packages made it faster
- Using simulation instead of an analytical solution most of the cases
 - Closed form solution for most systems are not possible

Areas of Application

- Manufacturing applications
- Semiconductor manufacturing
- Construction of engineering and project management
- Military applications
- Logistics, supply chain, and distribution applications
- Transportation modes and traffic
- Business process simulation
- Health care

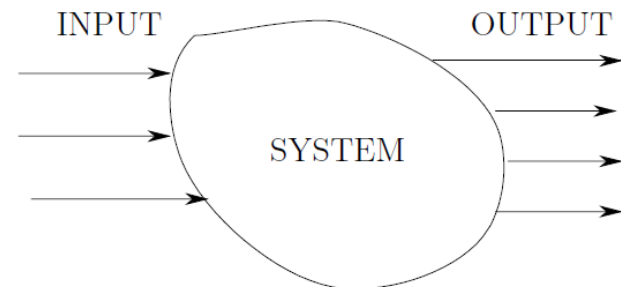
Systems – Definition

Qualitative Definition

- An aggregation of things to form an integral or complex whole (Encyclopedia of Americana).
- A regularly interacting or interdependent group of items forming a unified whole (Webster's Dictionary)
- A combination of components acting together to perform a function not possible with any of the individual parts (IEEE Standard Dictionary).
- Salient Features
 - A system consists of interacting components
 - A system is associated with a function that it performs

Quantitative Definition

- A quantitative definition associates a set of measurable variables with a system.
 - By measuring them over time $[t_0, t_f]$, the system behavior can be described
- Input variables: set of time function that can be controlled
$$\mathbf{u}(\mathbf{t}) = [u_1(t), \dots, u_p(t)]^T \quad t_0 < t < t_f$$
- Output variables: set of variables representing the quantity of interest
$$\mathbf{y}(\mathbf{t}) = [y_1(t), \dots, y_m(t)]^T \quad t_0 < t < t_f$$



Systems: Input-Output Model

- The quantitative definition of a system represents a model of the system
- A model duplicates the behavior of a system
 - Mathematical relationship between input and output
- A set of function defines the input-output relationship

$$y_1(t) = g_1(u_1(t), \dots, u_p(t)), \dots, y_m(t) = g_m(u_1(t), \dots, u_p(t))$$

- The system model in mathematical form

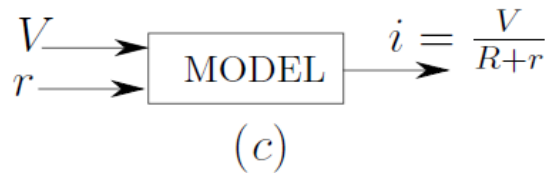
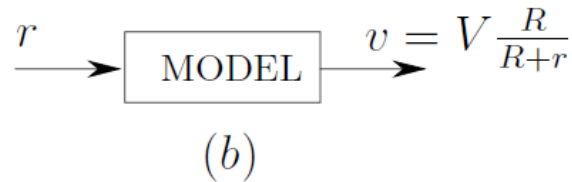
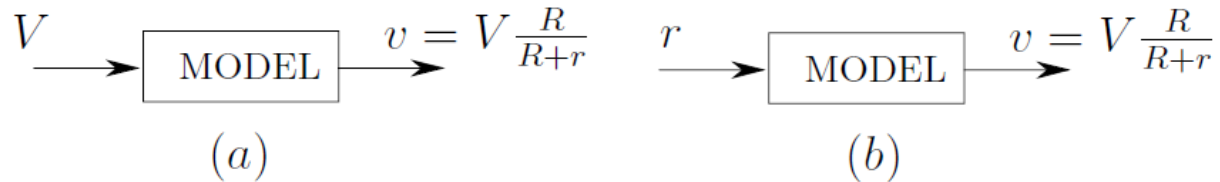
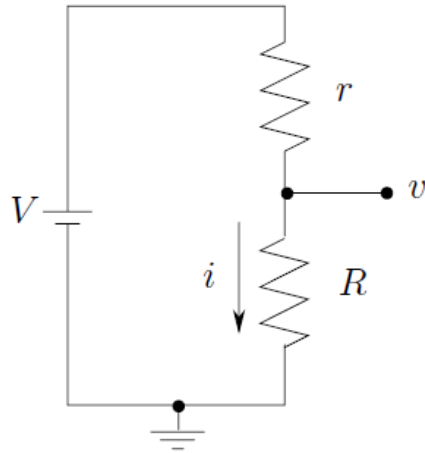
$$\mathbf{y} = \mathbf{g}(\mathbf{u}) = [g_1(u_1(t), \dots, u_p(t)), \dots, y_m(t) = g_m(u_1(t), \dots, u_p(t))]^T$$

Where $\mathbf{g}(\cdot)$ denotes the column vector with entities $g_1(\cdot)$, $g_m(\cdot)$.

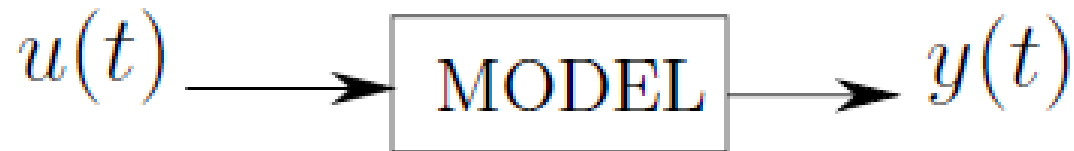
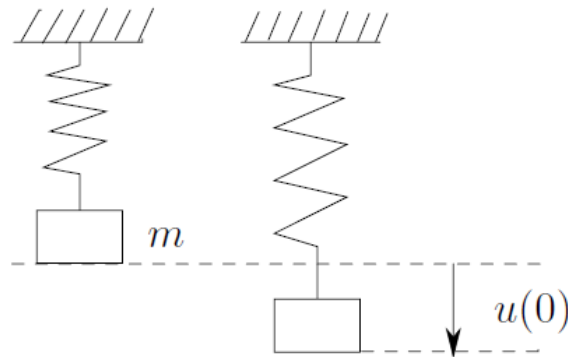


This is the simplest possible modeling process – Input-output modeling

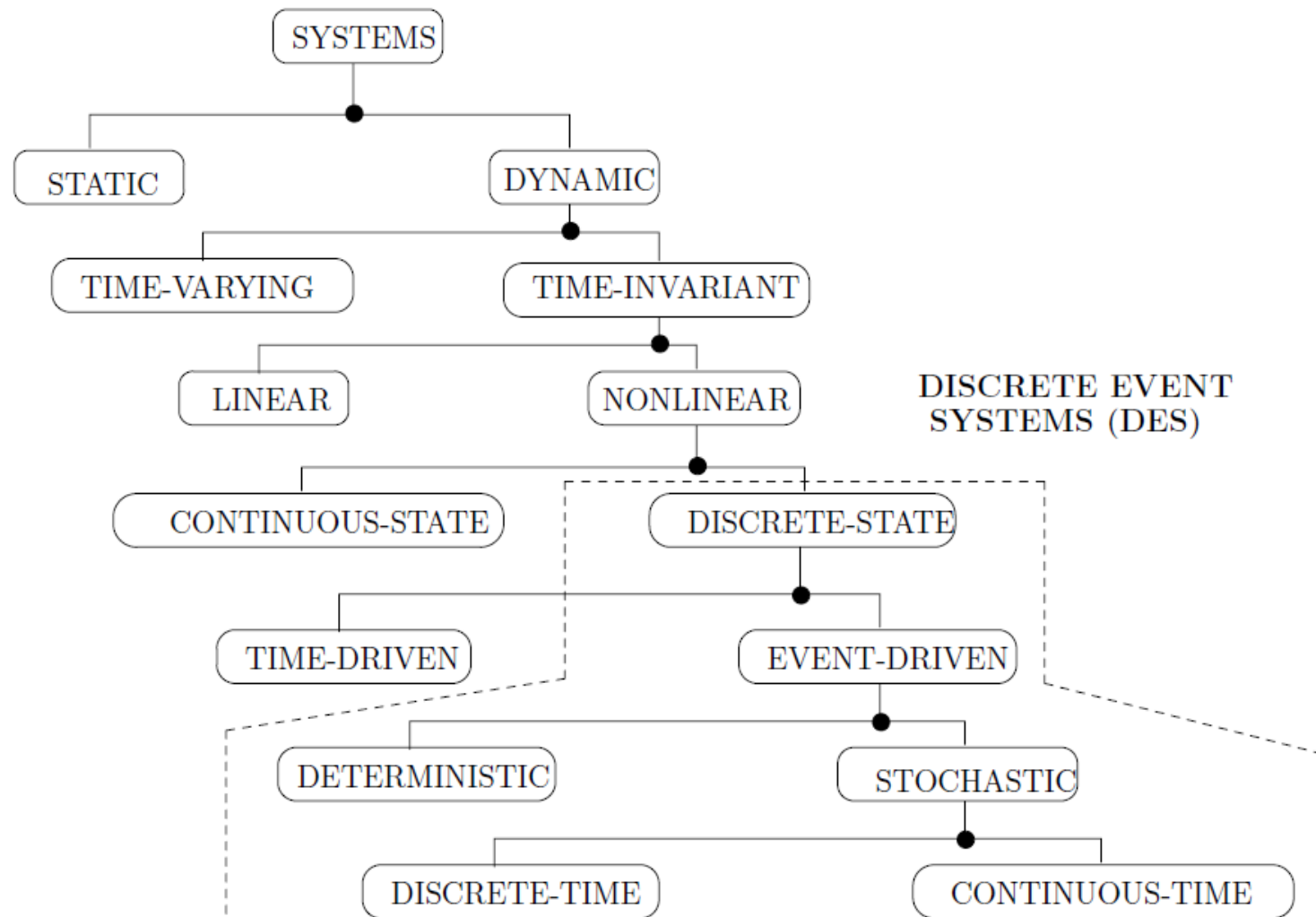
System Model – Example



System Model – Example (Contd.)



Taxonomy of System Classification



Static vs. Dynamic System Models

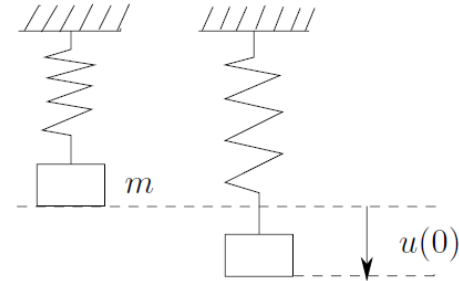
- Static System model
 - Time does not play a role
 - Input-output relationship are described by algebraic equations
 - $y(t)$ is independent of past input values
 - No memory is required
- Dynamic System model
 - It evolves over time
 - Input-output relationship is described by differential equation
 - $y(t)$ generally depends on past values of input

Input-Output Model – Dynamic Systems

- The Input-Output model is unable to predict the future output $y(t)$ given the $u(t)$ for a dynamic system
 - Output is a direct function of the current input
- Output of dynamic systems is affected by previous inputs
 - The impact of the past values should be stored in a set of variables
 - The values of these variables will represent the system state at that time

Model of a Dynamic System

- If at t_1 , the displacement is $y(t_1)$,
 - $y(t_1 + \tau) = ?$
 - Can we uniquely determine $y(t_1 + \tau)$ for $\tau > 0$
- There are three possibilities
 - $y(t_1 + \tau) > y(t_1)$, the mass moving downward
 - $y(t_1 + \tau) < y(t_1)$, the mass moving upward
 - $y(t_1 + \tau) = y(t_1)$, the mass is at rest
- How can we decide one of the above
 - What information is missing here
 - The velocity of the mass
- The input cannot only determine the output



The concept of State

- State of a system at any time describes the behavior of the system at that time in some measureable way
- The state of a system at time t is the information required at t to uniquely determine the output $\mathbf{y}(t)$, for all $t \geq t_0$, given $\mathbf{u}(t)$, for $t \geq t_0$

State Space Modeling

- In addition to input and output variables, a set of state variables, $\mathbf{x}(t)$, is defined
- Models represent the relationship among $\mathbf{u}(t)$, $\mathbf{y}(t)$, and $\mathbf{x}(t)$.
 - Reflects the system dynamics
- The set of equations required to specify the state $\mathbf{x}(t)$, for all $t \geq t_0$, given $\mathbf{x}(t_0)$ and the function $\mathbf{u}(t)$, $t \geq t_0$, are called state equations
- The state space of a system, X , is the set of all possible values that the state may take.

State Space Modeling (Contd.)

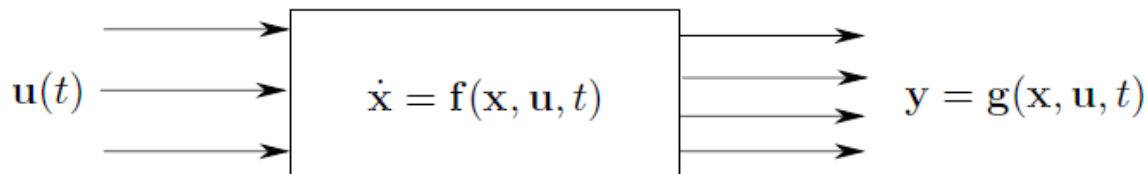
- State equations of a system usually depend on the system, however, it might have the following form

$$\dot{x}(t) = f(x(t), u(t), t)$$

- The state space model needs to specify the following set of equations

$$\dot{x}(t) = f(x(t), u(t), t), \quad x(t_0) = x_0$$

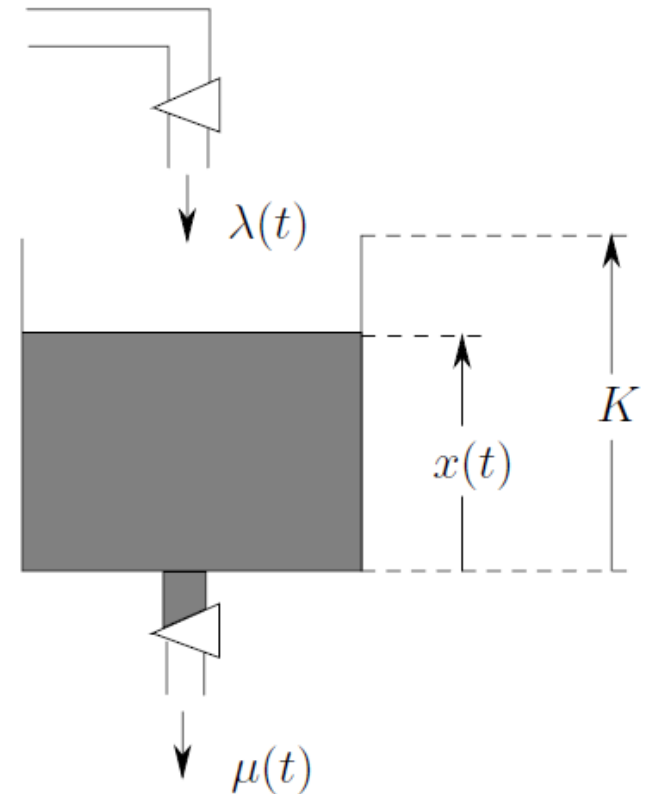
$$y(t) = g(x(t), u(t), t)$$



State Space Model: Example

- $\lambda(t)$ – rate at which flow in
- $\mu(t)$ – rate at which flow out
- $x(t)$ – desirable fluid level, and the state variable
- $0 \leq x(t) \leq K, t \geq 0$
- Complete state equation

$$\dot{x}(t) = \begin{cases} 0 & \text{if } [x(t) = 0 \text{ and } \lambda(t) \leq \mu(t)] \\ & \text{or } [x(t) = K \text{ and } \lambda(t) \geq \mu(t)] \\ \lambda(t) - \mu(t) & \text{otherwise} \end{cases}$$

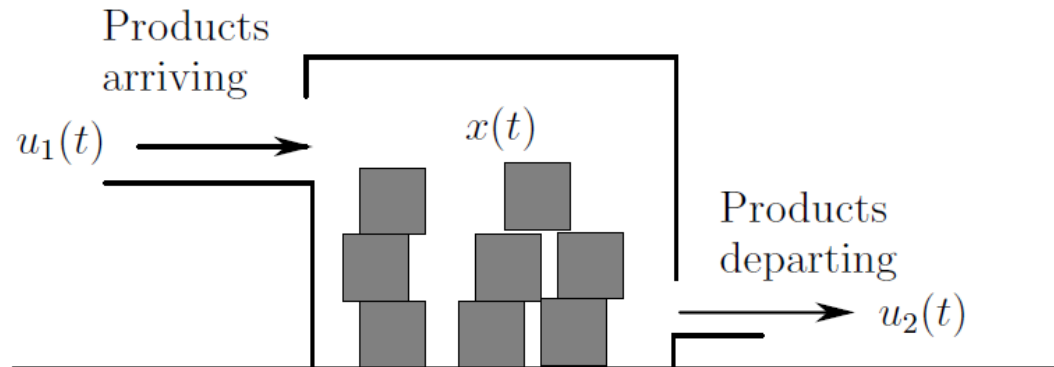


- The complete model is
 - $u_1(t) = \lambda(t)$ and $u_2(t) = \mu(t)$ are the input variables
 - $y(t) = x(t)$ is the output variable

Continuous- vs. Discrete-state systems

- Continuous-State System
 - The state space X is a continuous set
 - State equations are representing by differential equation
 - Mathematical tools and solution techniques are available
- Discrete-State System
 - The state space X is a discrete set
 - State equations are usually logical statements
 - If something happens and current state is x , next state will be x'
 - Mathematical representation and solving them are usually complex

Discrete-State System: Example



- State variable is $x(t)$
- State space is a set of non-negative integer, $X = \{0, 1, 2, \dots\}$
- Output equation, $y(t) = x(t)$
- Inputs are function of time defined as

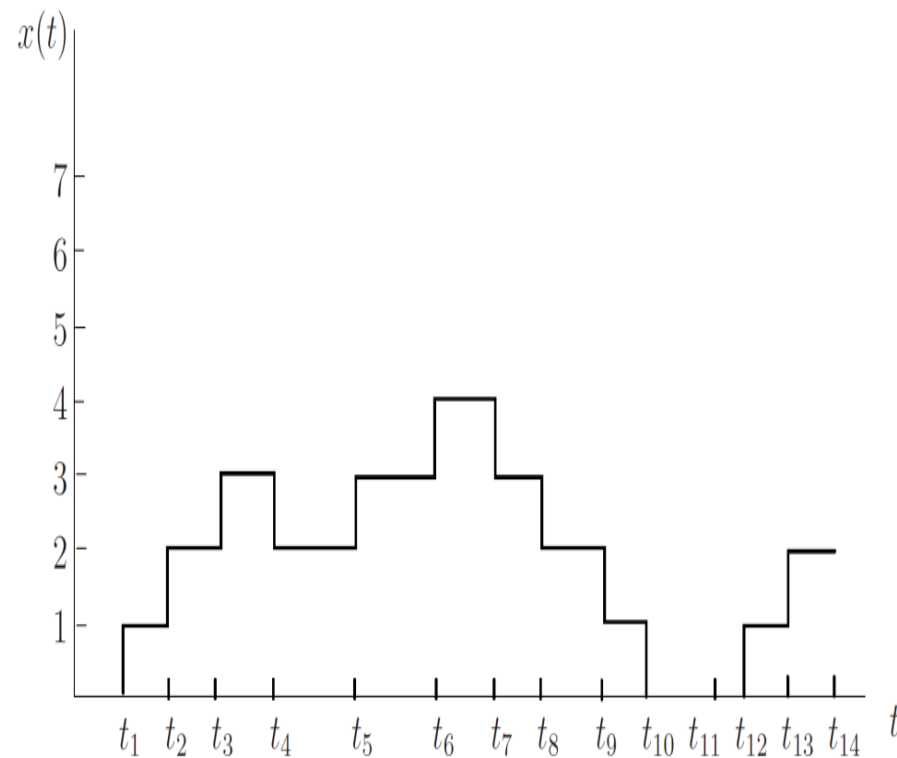
$$u_1(t) = \begin{cases} 1 & \text{if product arrives at time } t \\ 0 & \text{otherwise} \end{cases} \quad u_2(t) = \begin{cases} 1 & \text{if truck arrives at time } t \\ 0 & \text{otherwise} \end{cases}$$

Discrete-State System: Example (Contd.)

- State transitions of the system are
 - $u_1(t) = 1, u_2(t) = 0 \quad \gg \quad x(t) = x(t) + 1$
 - $u_1(t) = 0, u_2(t) = 1 \quad \gg \quad x(t) = x(t) - 1$
 - $u_1(t) = 0, u_2(t) = 0 \quad \gg \quad x(t) = x(t)$
- State equation: derivative is not possible. Let t^+ denote the time instant just after t .

$$x(t^+) = \begin{cases} x(t) + 1 & \text{if } (u_1(t) = 1, u_2(t) = 0) \\ x(t) - 1 & \text{if } (u_1(t) = 0, u_2(t) = 1, x(t) > 0) \\ x(t) & \text{otherwise} \end{cases}$$

Discrete-State System: Example (Contd.)



- Product arrives (i.e., $u_1(t) = 1$, at $t_1, t_2, t_3, t_5, t_6, t_{12}$ and t_{13}
- Truck arrives (i.e., $u_2(t) = 1$, at t_4, t_7, t_8, t_9 , and t_{10}

Concept of Events

- State space is described by a discrete set like $\{0,1,2, \dots\}$
- State transitions are only observed at discrete points in time
 - Transitions are associated with special actions
 - Such actions are termed as ‘Events’
 - Somebody Presses a button
 - Computer goes down
- Usually state transitions are associated with events

Discrete-event Simulation

- A discrete-event system is defined by
 - Stochastic – at least some of the system state variables are stochastic
 - Dynamic – the evolution of the system state
 - Discrete-event – system state is a discrete set and change in state is associated with events

Properties of Discrete Event Systems

- The state space is a discrete set
- The state transition mechanism is event-driven
- State evolution depends entirely on the occurrence of asynchronous discrete events
- The sample path of the system jumps from one state to another whenever an event occurs
- System status can be represented as sequence of timed events.

$(e_1, t_1), (e_2, t_2), (e_3, t_3), (e_4, t_4), (e_5, t_5), (e_6, t_6), (e_7, t_7)$

Examples of Discrete Event Systems

- Queueing systems
- Computer systems
- Communication systems
- Manufacturing systems
- Inventory systems
- Traffic systems
- Database systems

Thank You